🏆 Infosys SE Elite Crack Vault — 02 Multiplication Division Factor and Exponential Puzzles Elite

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Curation Standard: 100% Pure SE Role Specific** (All SP/DSE Upper-Level & Practice Variants Purged)
> **Total Curated Questions in Topic: 10 Questions**
> **Question Breakdown:** 🔴 `3 Must-Solve` | 🟡 `7 Core SE Level`
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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Question 1: Matrix Number Puzzle 16 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	3	2
64	8	8
63	9	?

Options

- ☐ **A)** 17
- ☐ **B)** 10
- ☑ **C)** 7
- ☐ **D)** 25

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 7**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 63 ÷ 9 = 7.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 7$.

4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

Distractor & Elimination Analysis

- **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.

Question 2: Matrix Number Puzzle 17 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	3	5
36	6	6
8	2	?

Options

- [ ] **A)** 12
- [ ] **B)** 8
- [ ] **C)** 19
- [x] **D)** 4

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option D) 4**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 2 = 4.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 4$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4.

Distractor & Elimination Analysis

- **Option A (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (8):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 3: Matrix Number Puzzle 18 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	2	3
8	4	2
72	8	?

Options

- [ ] **A)** 6
- [ ] **B)** 13
- [x] **C)** 9
- [ ] **D)** 3

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 9**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 72 ÷ 8 = 9.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 9$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

- **Option D (3):** Incorrect. Violates the established matrix row/column transformation rule.

Question 4: Matrix Number Puzzle 19 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	6	2
36	6	6
8	4	?

Options

- [x] **A)** 2
- [ ] **B)** 6
- [ ] **C)** 5
- [ ] **D)** 7

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option A) 2**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 4 = 2.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 2$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 2.

Distractor & Elimination Analysis

- **Option A (2):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (5):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

Question 5: Matrix Number Puzzle 20 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	5	3
49	7	7
42	6	?

Options

- [ ] **A)** 22
- [x] **B)** 7
- [ ] **C)** 12
- [ ] **D)** 4

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 7**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 42 ÷ 6 = 7.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 7$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

Distractor & Elimination Analysis

- **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Incorrect. Violates the established matrix row/column transformation rule.

Question 6: Matrix Number Puzzle 21 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	3	28
5	1	26
1	7	?

Options

- [x] **A)** 8
- [ ] **B)** 14
- [ ] **C)** 18
- [ ] **D)** 12

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option A) 8**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 1^2 + 7 = 8.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 8$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 8.

Distractor & Elimination Analysis

- **Option A (8):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (14):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (12):** Incorrect. Violates the established matrix row/column transformation rule.

Question 7: Matrix Number Puzzle 22 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	5	54
6	10	46
7	8	?

Options

- ☐ **A)** 49
- ☐ **B)** 54
- ☑ **C)** 57
- ☐ **D)** 45

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 57**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 8 = 57.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 57$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 57.

Distractor & Elimination Analysis

- **Option A (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (57):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

Question 8: Matrix Number Puzzle 23 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	5	41
6	1	37
5	2	?

Options

- ☐ **A)** 37
- ☐ **B)** 12
- ☑ **C)** 27
- ☐ **D)** 42

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option C) 27**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 27$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

Distractor & Elimination Analysis

- **Option A (37):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (42):** Incorrect. Violates the established matrix row/column transformation rule.

Question 9: Matrix Number Puzzle 24 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
3	5	14
5	4	29
5	2	?

Options

- ☐ **A)** 15
- ☑ **B)** 27
- ☐ **C)** 21
- ☐ **D)** 23

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 27**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 27$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

Distractor & Elimination Analysis

- **Option A (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (23):** Incorrect. Violates the established matrix row/column transformation rule.

Question 10: Matrix Number Puzzle 25 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
1	2	3
6	5	41
7	5	?

Options

- ☐ **A)** 66
- ☑ **B)** 54
- ☐ **C)** 49
- ☐ **D)** 39

Correct Answer

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: Option B) 54**

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 5 = 54.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 54$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 54.

Distractor & Elimination Analysis

- **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (39):** Incorrect. Violates the established matrix row/column transformation rule.
